Lecture 5

What do I do with a list of genes?

Functional Analyses

We are jumping WAY ahead here

We are showing you the "future".

It is good to know where you are going.

Most bioinformatics analysis produces the type of results that we discuss today.

These skills are useful even if you don't need to do the final interpretation yourself.

What your results will look like

A surprising number of "final" data analysis results will be on of:

1. A **list of names**
2. A **list of names** with a **single value**
3. A **list of names** with a **matrix of values**

The names may be **gene**, **transcript** or other **feature names**.

The question becomes to how do you interpret the list?

A list of genes

See the online lecture information for a link to the files. Download the `gene-list.txt` file with `wget` then:

```
cat gene-list.txt | head -5
```

will be:

```
ACE
ADRB2
ADRB3
AGRP
AKR1C2
```

The list is limited to a *small subset* of all names!

A list of genes with values

Displays a name with a single measurement:

```
cat gene-values.txt | head -5
```

Prints:

```
gene        value
Tmem132a    1.04E-12
Myl3        6.67E-14
Myl4        3.27E-09
Hspb7       1.27E-07
```

Every gene name is present, the values for each are different.

A gene matrix

Displays a name with multiple measurements

```
cat gene-matrix.txt | head -5
```

Prints

```
gene      C1      C2       C3       M1      M2      M3
Tmem132a  7349.5  10604.4  11400.6  694.7   709.3   760.2
Myl3      1207.1  1345.0   1247.6   2222.9  3041.3  2819.0
Myl4      2468.6  2588.5   2840.4   3963.2  5044.9  4824.7
Hspb7     562.5   610.3    647.0    947.0   1300.0  1144.7
```

Every gene name is present, the values are different.

But what does it all mean?

The Known Knowns

First you ought to understand:

---

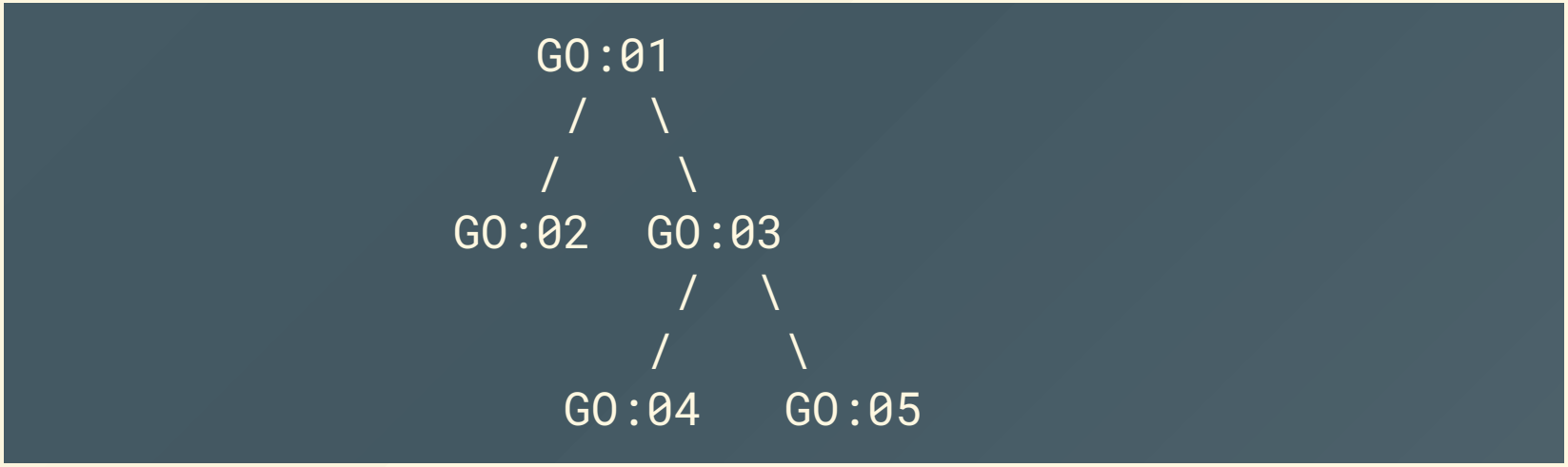
1. How is the current knowledge represented.
2. How is the current knowledge searched.

Then go on and search your data against this knowledge now knowing what you could expect to get back.

The Gene Ontology

What is the GO?

- Definition of GO Terms layed out as a tree:



- Association of gene products with terms.

```
Gene  A     GO:03
```

Let's investigate the GO data

The gene ontology definition file (detailed commands in the book)

```
wget http://purl.obolibrary.org/obo/go.obo
```

how many lines:

```
cat go.obo | wc -l
# 632140
```

Page through it with

```
more go.obo
```

GO Terms

The content of the core `go.obo` file is constructed of records in the form:

```
[Term]
id: GO:0000002
name: mitochondrial genome maintenance
namespace: biological_process
def: "The maintenance of the structure and integrity of
  the mitochondrial genome; includes replication and
  segregation of the mitochondrial chromosome." [GOC:ai, G
is_a: GO:0007005 ! mitochondrion organization
```

The `is_a` line indicates the *parent* of the term.
If your "concept" is not in this file tools will not find it.

Manipulate GO files from command line

Once you identify the patterns in the records, then you can search for various content:

```
cat go.obo | grep "namespace: biological_process" | wc -l
# 30583

cat go.obo | grep "namespace: molecular_function" | wc -l
# 12123

cat go.obo | grep "namespace: cellular_component" | wc -l
# 4300
```

Every functional enrichment tool uses this file as its basis.

Search the file for functions

You can also get previous and following lines by passing the `-B` (before) `-A` (after) options to `grep`.

```
cat go.obo | grep "lactase activity" -B 2 -A 5 | head -8
```

Prints:

```
[Term]
id: GO:0000016
name: lactase activity
namespace: molecular_function
def: "Catalysis of the reaction: lactose + H2O = D-glucose
synonym: "lactase-phlorizin hydrolase activity" BROAD [EC:
synonym: "lactose galactohydrolase activity" EXACT [EC:3.2
xref: EC:3.2.1.108
```

The association file for Homo Sapiens

From the GO download page, copy the link then:

```
wget http://geneontology.org/gene-associations/goa_human.g

# Unzip the compressed file.
gunzip goa_human.gaf.gz
```

How big is the resulting file:

```
cat goa_human.gaf | wc -l
# 425901
```

There you have it. `425,901` known functions for the human genes.

What is in the association file?

There is a readme with the file (on the web) and you can download that the same way.

You can also page through the file

```
cat goa_human.gaf | more
```

Comments are specfied with `!` the rest are tab separated and column oriented data.

Remove the lines starting with `!` to simplify it.

```
cat goa_human.gaf | grep -v '!' > assoc.txt
```

What properties do the data have?

The GAF format states that column 3 has to be a

> *a symbol that means something to a biologist wherever possible (a gene symbol, for example)*

```
cat assoc.txt | cut -f 3 | head
```

Prints

```
DNAJC25-GNG10
DNAJC25-GNG10
DNAJC25-GNG10
HDGFRP3
HDGFRP3
```

How many gene symbols?

```
cat assoc.txt | cut -f 3 | sort | uniq -c | wc -l
# 19421
```

Most genes appear to have at least one entry. `425,901` over `19,421` genes means on average about `22` annotation per gene.

But the annotations are not evenly distributed.

Annotation distribution

Redirect output into `gene_counts.txt`:

```
cat assoc.txt | cut -f 3 | sort | uniq -c | sort -k1,1nr >
cat gene_counts.txt | head
```

The "top" genes have annotations way above the `22`

```
724 TP53
669 GRB2
637 EGFR
637 UBC
580 RPS27A
570 UBB
565 UBA52
511 CTNNB1
422 SRC
```

Command line analytics with datamash

A handy tool called `datamash` lets you do data analytics at command line.

```
# Activate your enviroment
source activate bioinfo

# Get help on datamash
datamash --help
```

Unfortunately the `uniq -c` command pads numbers with a variable number of spaces. We need to `squeeze` those into a single space. `tr -s` can do that.

```
cat gene_counts.txt | tr -s ' '
```

Working with datamash

Average number annotations:

```
cat gene_counts.txt | tr -s ' ' | datamash -t ' ' mean 2
# 21.928170537048
```

You can list multiple operations at a time:

```
cat gene_counts.txt | tr -s ' ' | datamash -t ' ' mean 2
# 21.928170537048 1 724
```

Enrichment analysis

Given a selected set of genes and their annotations are there functional roles common to most of these genes?

Enrichment analysis answers this question. It is typically one of the last steps of the analysis. It is about making sense of the results.

Best if done by a *domain expert*.

There are many tools to do enrichment – may produce different results.

Overrepresentation analysis

The book shows several options (many more exists):

- **g:Profiler**
- **Panther**
- **DAVID**
- **ermineJ**

Tools come with different tradeoffs and my better suited for different problem sets.

It is not clear beforehand which tool works for a given problem.

Will different tools produce different results?

Biostar Quote of the Day: Why does each GO enrichment method give different results?

> *I'm new to GO terms. In the beginning it was fun, as long as I stuck to one algorithm. But then I found that there are many out there, each with its own advantages and caveats (the quality of graphic representation, for instance). [...] As a biologist, what should I trust? Deciding on this or that algorithm may change the whole story!*

Data selection for the lecture

If you are pursuing a research project already you may have your own data available. Use that when you follow the examples in the book.

You can also make your own data. Example: take the top 20 most annotated genes from GO and see what is common about them.

You may also download the `gene-list.txt` from the lecture website.

The same applies to the homework.

Explore the options.

Know what could be done.

The book has many examples.